Recent Advances in Pure, Industrial and Applied Mathematics driven by AI and Machine Learning - 2025

On the $k-$Domination Vertex Connectivity of Connected Graphs

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5-6 June, 2025

Objective

Find the $k-$domination vertex connectivity for connected graphs like Path graphs, Cycle graphs and Complete graphs.

Vertex Connectivity

- Kenneth Rossen defined the concept of vertex connectivity in graphs [4] as:

Definition

The minimum number of vertices whose deletion makes the graph G disconnected is called the **vertex connectivity** of a connected graph, denoted by $\gamma(G)$.

- $\gamma(G)$ is the size of the smallest subset of separating set.

[4] Kenneth H. Rossen, Discrete Mathematics and its applications, *Mc Graw Hill*, 7th ed., 2012, pp. 678-685.

Example: Vertex Connectivity

- Let us consider a connected graph G.

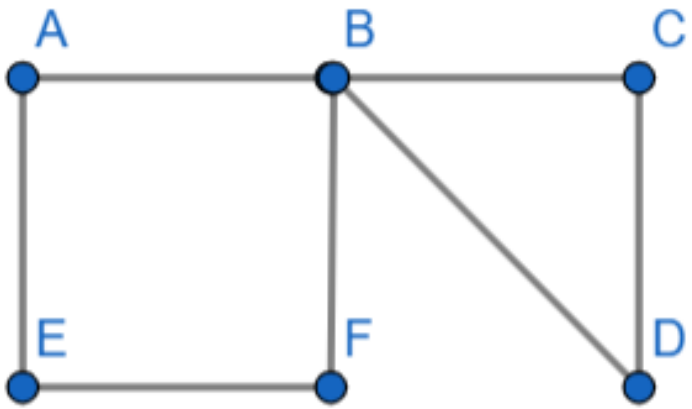

Figure 1: G

Figure 2: G_1

- From Figure 1, if we remove the vertex B then it will disconnect as shown in Figure 2.
- The graph can be disconnected in many ways:
  $S = \{\{B\}, \{B, D\}, \{B, D, F, \}, ...\}$ is the separating set.
- The cardinality of the minimum subset of S is the vertex connectivity of the graph G, $\gamma(G) = 1$.

Edge Connectivity

- R. Balakrishnan and K. Ranganathan defined the concept of edge connectivity in graphs [2] as:

Definition

The minimum number of edges whose deletion makes the graph G disconnected, is called the **edge connectivity** of a connected graph, denoted by $\lambda(G)$.

- $\lambda(G)$ is the cardinality of the smallest subset of the disconnecting set.

[2] R. Balakrishnan, K. Ranganathan. A textbook of Graph Theory, *Springer, Universitext, New York*, 2nd ed., 2012, pp. 49-59, 212-228.

Example: Edge Connectivity

- Let us consider a connected graph G.

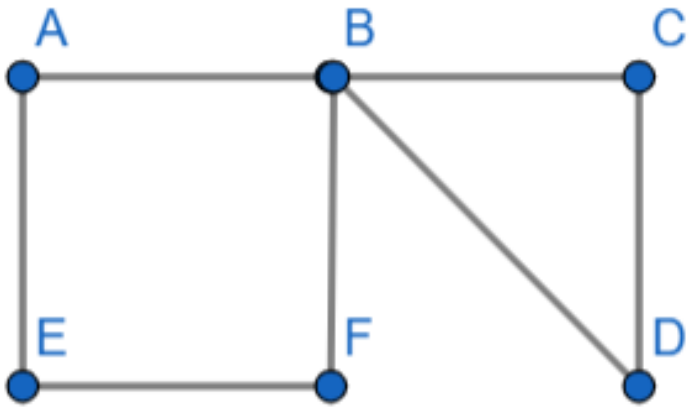

Figure 3: G

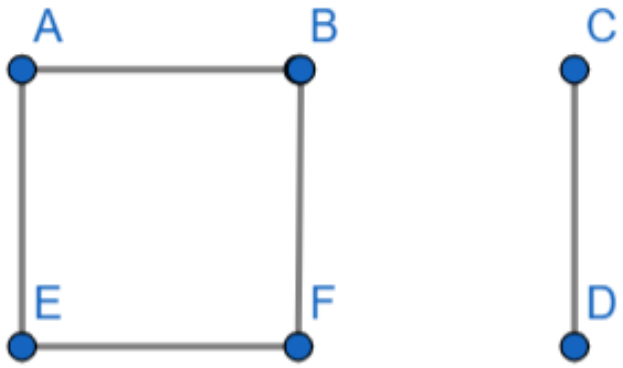

Figure 4: G_2

- After removing two edges {{BC,BD} from Figure 3 it disconnects the graph G, shown in figure 4.
- The graph can be disconnected in many ways: $T = \{\{AB, BF\}, \{BC, BD\}, \{AB, BC, BF, BD\}, ...\}$ is the disconnecting set.
- The cardinality of the minimum subset of T is the edge connectivity of the graph G, $\lambda(G) = 2$.

Domination number

- In the year 1996 [3], R. J. Wilson introduced the concept of dominating set and domination number as :

Definition

A **dominating set** D for a graph G is a subset of vertex set of G, such that any vertex of G is either in D or has a neighbour in D.

Definition

The **domination number**, denoted by $k(G)$, it is the cardinality of the smallest dominating set of graph G.

[3] Robin J. Wilson, Introduction to Graph Theory, *Addison Wesley Longman Limited, England*, 4th ed., 1996, pp. 26-30.

Example: Domination number of a Path

- Let us consider a path graph P_5, on 5 vertices, as shown in the Figure 5.

Figure 5: Domination number of a path

- Here, $D_1 = \{B, D\}$ and $D_2 = \{A, C, E\}$ are the dominating sets of P_5.
- The cardinality of the set D_1 is 2, which is smallest amoung all dominating sets. Thus, the domination number of P_5 is $k(P_5) = 2$.

Example: Domination number of a cycle

- Similarly, consider a cycle C_5 on 5 vertices.

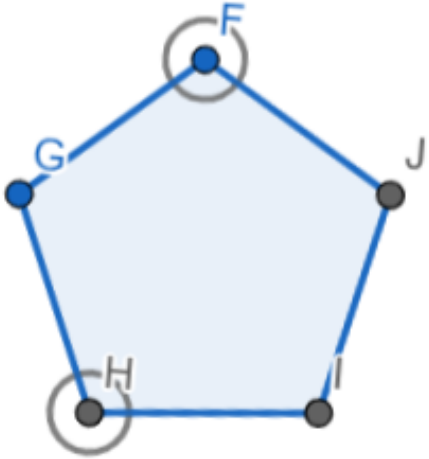

Figure 6: Domination number of a cycle

- $D_1 = \{F, H\}$ is the dominating set of the cyclic graph.
- Hence, the domination number of C_5 is $k(C_5) = 2$.

k-Domination Edge Connectivity

- In the paper [1], the authors introduced the term k-domination edge connectivity number, defined as:

Definition

Let $G = (V, E)$ be a connected graph where V is the **vertex set**, E is the **edge set** of G and E' be any non-empty subset of E. If $G - E'$ is disconnected and every disconnected component has domination number equal to k, then the cardinality of the smallest subset of E' is called the k-**domination edge connectivity number** of G and it is denoted by $\lambda^{(k)}(G)$.

[1] M. Serif Aldemir, Suleyman Ediz, Idris Ciftci, Kerem Yamac, Ziyattin Tas. Domination Edge Connectivity, *Graphs and Linear algebra*, (2), 2024, pp. 1-10.

Example: k-Domination Edge Connectivity

- Let us consider a path graph on 9 vertices, denoted by P_9, as shown in the Figure 7.

Figure 7: Path with 9 vertices

- The set $S = \{EF\}$ is the edge-disconnecting set such that on disconnecting the path graph into 2 components, each component has the domination number, $k = 2$, as shown in the figure 8.

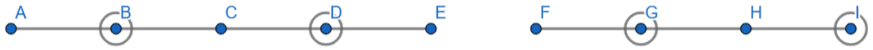

Figure 8: 2-domination edge connectivity

- Hence, the 2-domination edge connectivity, $\lambda^{(2)}(P_9) = 1$.

k-Domination Vertex Connectivity

- Similarly, we have introduced the concept of k-Domination Vertex Connectivity as:

Definition

Let $G = (V, E)$ be a connected graph where V is the **vertex set**, E is the **edge set** and V' be any non-empty subset of V. If $G - V'$ is disconnected and every disconnected component has domination number equal to k, then the cardinality of the smallest subset of V' is called the k-**domination vertex connectivity** of G and it is denoted by $\gamma^{(k)}(G)$.

Example: k-Domination Vertex Connectivity

- Let us consider a path graph with 9 vertices, denoted by P_9, as shown in the Figure 9.

Figure 9: Path with 9 vertices

- The set $T = \{E\}$ is the vertex-disconnecting set such that on disconnecting the path graph into 2 components, each component has the domination number, $k = 2$, as shown in the figure 10.

Figure 10: 2-domination vertex connectivity

- Hence, the 2-domination vertex connectivity, $\gamma^{(2)}(P_9) = 1$.

Results: Domination number

- In 2020, Casinillo [5], stated the theorem:

Theorem 1

The domination number of a path graph and a cycle on n vertices is $\lceil \frac{n}{3} \rceil$.

[5] Casinillo Leomarich, New Counting Formula for Dominating sets in Path and Cycle, *Journal of Fundamental Mathematics and Applications*, November 2020, pp. 1-4.

Results: Domination number

- Using **Theorem 1** from [5], we derived the following theorem.

Theorem 2

The domination number of a r-regular connected graph on n vertices is $\lceil \frac{n}{r+1} \rceil$.

- The degree of each vertex in a complete graph K_n is $n-1$. Thus we get the following corollary.

Corollary 3

The domination number of a complete graph K_n is 1.

[5] Casinillo Leomarich, New Counting Formula for Dominating sets in Path and Cycle, *Journal of Fundamental Mathematics and Applications*, November 2020, pp. 1-4.

Results on k-domination vertex connectivity of the graph.

- We aim to understand how the total number of vertices in a connected graph with domination number of the components k can be decomposed.

Theorem 4

Let k be the domination number of a component of a graph on n vertices. We can write $n = (3k-2)a + (3k-1)b + (3k)c + \gamma$, where a, b, and c are the number of components of $3k-2, 3k-1$, and $3k$ vertices respectively, and γ is the k-domination vertex connectivity of the graph.

- We now explore how the k-domination vertex connectivity γ relates to the number of connected components in the graph.

Proposition 5

Let γ be the k-domination vertex connectivity of a graph, then total number of components is either $\gamma + 1$ or γ.

Results: Combinations and Bounds

- Now, we consider how many different ways the graph can be split into components while preserving a domination number k and vertex connectivity γ.

Proposition 6

Let k be the domination number of each component of a graph with the vertex connectivity being γ, then total number of possible combinations of components are given by $k^{\gamma+1}$.

- In **Theorem 7** we have established an upper bound on the vertex connectivity in terms of total number of vertices and domination parameters.

Theorem 7

Let n be the total number of vertices of a graph with vertex connectivity γ and each component having domination number k, then $\gamma \leq \frac{n}{3k}$.
Equality holds only when the number of components is equal to γ.

Results: Minimum number of vertices

- We explore the relationship between the size of a connected path graph, its domination number, and the minimum vertices required to determine its vertex connectivity.

Theorem 8

Let n be the number of vertices of a connected path with domination number, k, then the minimum number of vertices required to find the $k-$ domination vertex connectivity, γ is $3(2k-1)$,

$$\text{i.e., } n \geq 3(2k-1).$$

- This theorem sets a critical lower bound on the graph size, ensuring sufficient vertices exist to accurately determine vertex connectivity based on domination number.

Result: 2-domination vertex connectivity

- We now focus on the vertex connectivity of a path graph P_n when the domination number is fixed at $k = 2$.

Theorem 9

Let n be the number of vertices of a path graph P_n, then the vertex connectivity with domination number, $k = 2$, is

1. 2-domination vertex connectivity, $\gamma^{(2)}(P_n) = \lfloor \frac{n-q+1}{6} \rfloor$, for $n = 6q + r$, $0 \leq r < 6$, and $q \neq r$.

2. 2-domination vertex connectivity, $\gamma^{(2)}(P_n) = q$, for $n = 6q + r$, $q = r$.

Result: 3-domination vertex connectivity

- Let us explore how the vertex connectivity behaves in a path graph P_n when its domination number is set to $k = 3$.

Theorem 10

Let n be the number of vertices of a path graph P_n, then the vertex connectivity with domination number, $k = 3$, is

1. 3-domination vertex connectivity, $\gamma^{(3)}(P_n) = \lfloor \frac{n-q}{9} \rfloor$, for $n = 9q + r$, where $r \neq (q-1)$ and $0 \leq r < 9$.

2. 3-domination vertex connectivity, $\gamma^{(3)}(P_n) = \lfloor \frac{n-q-1}{9} \rfloor$, for $n = 9q + (q-1)$.

Result: k-domination vertex connectivity

- We have generalized the expression for $k-$ domination vertex connectivity for path graphs and cycles for domination number $k > 3$.

Theorem 11

Let $n = (3k)q + r$, $0 \leq r \leq 3k$ and $n \geq (3k-2) * (k + \lfloor \frac{k}{2} \rfloor - 1) + (k + \lfloor \frac{k}{2} \rfloor - 2)$, with the domination number $k > 3$, the $k-$ domination vertex connectivity is $\gamma = \lfloor \frac{n-q}{3k} \rfloor$.

Code

- To automate the computation of vertex connectivity $\gamma^{(k)}(P_n)$ based on the derived formulas, we implemented the following code.

- It takes as input the total number of vertices n and the domination number k, and returns the corresponding $k-$ domination vertex connectivity.

Code

```python
#k-Domination vertex connectivity of a path graph

import math

n = int(input('Enter the number of vertices of a path graph, n: '))
k = int(input('Enter the domination number, k: '))
q = n // (3*k) #Quotient
r = n % (3*k) #Remainder
print(f'{n} = ({3*k}){q} + {r}')

if n >= 3*(2*k - 1):
  print(f'Minimum number of vertices required to find the {k}-domination vertex connectivity is satisfied.')

  print(f'{k}-domination vertex connectivity of a path graph with {n} vertices is, gamma <= {math.ceil(n/(3*k))}.')

  if k==2:
    if q != r:
      gamma = math.floor(( n - q + 1) / 6)
      print (f'{k}-domination vertex connectivity of a path graph of {n} vertices is {gamma}.')

    elif q == r:
      gamma = math.floor(( n - q) / 6)
      print (f'{k}-domination vertex connectivity of a path graph of {n} vertices is {gamma}.')

  elif k == 3:
    if r != q-1:
      gamma = math.floor(( n - q) / 9)
      print (f'{k}-domination vertex connectivity of a path graph of {n} vertices is {gamma}.')
```

Figure 11: Code1

Code

```
  elif k == 3:
    if r != q-1:
      gamma = math.floor(( n - q) / 9)
      print (f'{k}-domination vertex connectivity of a path graph of {n} vertices is {gamma}.')

    elif r == q-1:
      gamma = math.floor(( n - q - 1) / 9)
      print (f'{k}-domination vertex connectivity of a path graph of {n} vertices is {gamma}.')

  else:

    if n >= (3*k - 2) * (k + math.floor(k/2) - 1) + (k + math.floor(k/2) - 2):
      gamma = math.floor(( n - q) / (3*k))
      print (f'{k}-domination vertex connectivity of a path graph of {n} vertices is {gamma}.')

    elif n < (3*k - 2) * (k + math.floor(k/2) - 1) + (k + math.floor(k/2) - 2) and n >= 3*(2*k-1) :
      gamma1 = q + r
      gamma2 = math.floor(( n - q) / (3*k))
      print (f'{k}-domination vertex connectivity of a path graph of {n} vertices is either {gamma1} or {gamma2}.')

  print(f'The general expression of number of vertices, domination number, and k-domination vertex connectivity is: {n} = ({3*k - 2})a + ({3*k - 1})b + ({3*k})c + {gamma}.')

  print(f'Total number of components with the {k}-domination vertex connectivity being {gamma} for a path graph of {n} vertices is either {gamma} or {gamma + 1}.')

  print(f'Total number of possible combinations of components for the {k}-domination vertex connectivity of a path graph being {gamma}, is {k**(gamma+1)}.')

else:
  print(f'Minimum number of vertices required to find the {k}-domination vertex connectivity is not satisfied.')
```

Figure 12: Code2

Further Research

1. In Theorem 4, we will find the number of components a, b, and c of $3k-2, 3k-1$, and $3k$ vertices respectively.

2. From Theorem 8 and Theorem 11, we get the k-domination vertex connectivity for $(3k-2)*(k+\lfloor\frac{k}{2}\rfloor-1)+(k+\lfloor\frac{k}{2}\rfloor-2)$.
   Now, we have found that for
   $3(2k-1) < n < (3k-2)*(k+\lfloor\frac{k}{2}\rfloor-1)+(k+\lfloor\frac{k}{2}\rfloor-2)$, the k-domination vertex connectivity is either $\gamma = \lfloor\frac{n-q}{3k}\rfloor$ or $\gamma = q+r$, where $n = (3k)q+r, 0 \leq r \leq 3k$.
   Further, we will work on finding the suitable k-domination vertex connectivity formula of the above two equations, which will satisfy for each n.

3. We will find the k-domination vertex connectivity for graphs like Trees, Cycles, Regular and Complete graphs.

Conclusion

The study on k-Domination Vertex Connectivity provides a deeper understanding of how the vertex connectivity number, γ, is influenced by the domination number,k, across various types of graphs, such as paths, cycles, regular graphs, and complete graphs. The results highlight key equations and bounds that establish relationships between the number of vertices, vertex connectivity, and the domination number.
Significant findings include:

1. The formulation of general equations connecting γ and k for different graph types.
2. Identification of conditions and bounds for vertex connectivity concerning domination numbers.
3. Detailed insights into 2-domination, 3-domination, and k-domination vertex connectivity.

Future work will focus on refining formulas for k-domination vertex connectivity and extending the analysis to additional graph classes such as trees and complete graphs.

References

- M. Serif Aldemir, Suleyman Ediz, Idris Ciftci, Kerem Yamac, Ziyattin Tas. Domination Edge Connectivity, *Graphs and Linear algebra*, (2), 2024, pp. 1-10.
- R. Balakrishnan, K. Ranganathan. A textbook of Graph Theory, *Springer, Universitext, New York*, 2nd ed., 2012, pp. 49-59, 212-228.
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Thank you!